InfoMorph System Architecture

Frontend / Browser

React 19 + TypeScript + TanStack Router

Entry Points

Story landing
Studio
Sign-in

Source Inputs Accepted

Pasted source text
Selectable text PDF
Readable web URL

PDF is Text Extraction (In Browser)

Extract text from selectable PDF's client-side

UI / Features

Drafts
Source facts
Claim evidence
Revision flags
Edit / re-verify
Manual export

Subbase Auth
sign-in / session

Supabase PodbaseSQL
authenticated save / reload

Private Subbase Storage
PDF / audio upload / download

Source text, **selected channels,** writing settings, URL or revised source

drafts, evidence,,(status, affectd-uped flags (validation / checks)

Backend / Application Server

TerStack Start server functions
Zod input / output validation
CSRF middleware

Server-Side Processing
URL fetch extracts readable HTML/ plain text

Analyze Source
extract facts and quotations

Generate Drafts
selected channel drafts

Verify Claims
check text claims

Compare Revisions
compare reoled sources

Server-Only Gemini Adapter
private server credential

pump! and source content (private server credentials)

strucured **AI response**

AI / Models and Configuration

Direct Gemini API
Text model selected via INFORMPHModel.
GEMINI_API_KEY side-side only

Gemini Tasks

Fad extraction with quotations
Content generation
Semantic claim comparison
Source revision comparison

Optional speech / video model overrides account support and live validation requirement

Supabase

Supabase Auth

Sessions
Email / password sign-in
Google logn via Lonsbo Cloud auth support

Saving requires sign-in; generation does not always.

Supabase PostreSQL

projects
source version
general content
revision, changes

Per user row-level security

Stores source versions, drafts, verification results, version records

Private Supabase Storage

source documents (original PDFs, generated media (audio only)

Use specific folder policies

Not rendered video clips persated

Document Processing Pipeline

1. Input text / PDF / URL
2. Extract text browser PDF js or server URL fetch
3. Analyze facts + quotations
4. Generate channel drafts
5. Verify AI comprison / determinic checks
6. Human review, export or authenticated save

User-triggered Revision Branch (not continuous monitoring)

1. User provides revised source
2. Compare old / new facts
3. Flag affected drafts
4. Regenerate and review

Verification checks source consistency; not independent truth.

Selectable PDFs: 10 MB and up to 50 pages, source text: 20-30,000 characters. No OGR.

Optional media and full deployed workflow require live validation.

Repository architecture baseline: ade0995.